S. Sowarna
Jyyappan

DA - 1.

21N181103

Q1) High Performance computing (HPC) clusters, load balancing clusters and stored clusters for VIT university.

Ans:

VIT university is a premier educational institutional known for its academic excellence & research contributions, can significantly enhance its operational efficiency, research capabilities and administrative functions through the implementation of various cluster computing systems, cluster computing involves connecting multiple computers to function as a unified system, offering advantages such as improved processing power, reliability and scalability.

High Performance cluster computing (HPC):

HPC clusters are essential for computing intensive tasks, making them ideal for VIT's research and academic needs.

These clusters can be implemented in following ways:

1. Research and Development support:

Many research & projects at VIT involves simultaneous, machine learning / AI & large scale data processing. An HPC cluster can provide researches with necessary computing power to run complex algos efficiently.

2. Parallel Programing

These clusters enable parallel processing, allowing multiple computations to be executed simultaneously. This enhances the speed and efficiency.

3) Integrating with curriculum:

By incorporating HPC clusters into various courses, students can gain hands on experience in high performance computing.

4) Loading Balance clusters:

Load Balancing clusters distribute workloads across multiple servers to optimize resources utilization and ensure uninterrupted access to university applications. VIT can implement load balancing clusters in following areas:

1. Online Learning Platforms:

With a significant no. of students accessing the learning management system (LMS), a load balancing clusters can ensure seamless performance during peak usage times.

2. FFCS Registration & Hostel Booking:

FFCS & other registration portals require high availability.

Load Balancing clusters can prevent server overloads and maintain accessibilities.

3. Libraries & Digital Resources:

This university library and digital repository require consistent uptime, which load balancing clusters can ensure.

Storage clusters:-

storage clients provides a scalable & fault tolerant data storage solution, which is crucial for managing large scale student records, research data and academic material.

VIT can implement storage clusters for

1: Student & Faculty Data management

Securely storing academic remarks, personal details & financial data.

2. Research Data Storage:

Large scale research projects generate vast amount of data that requires high capacity, redundant storage

systems.

3. Cloud Integrations:

combining storage clusters with cloud services allows remote access & additional availability options.

2.)

Question2: Private, Public, Hybrid and community cloud for VIT University.

cloud computing has revolutionized how educational institutions operate offering scalable, cost-effective and flexible computing resources. VIT university can leverage different cloud computing models to optimize it's academic, research & administrative functions effectively

Public cloud:

Public cloud are managed by third party providers and offer scalability and cost effectiveness. VIT can utilize public cloud service for,

Service for,

E - commerce Platforms: Hosting VITOL, online classes and virtual labs on public cloud platforms ensures global accessibility & minimal infrastructure costs.

2. collaboration tools:

cloud based tools like google workspace and microsoft 365 facilitate seamless communication & collaboration among students & faculty.

3. Data Analytics & AI services:

Leveraging AI and Data Analytics on public cloud services enables advanced research applications in various fields.

* Private cloud:

A private cloud offers enhanced security and control, making it ideal for handling sensitive university data.

VIT can implement private cloud solutions for :

1. Secure student & staff Data :

Academic records, payroll systems and confidential research data can be stored securely on private cloud.

2. Cloud Applications :

Hosting internal applications with ERP systems examination portal & attendance tracking system ensures data security and accessibility.

3. High Performance computing :

Research project requiring heavy computational power can be executed efficiently within a private cloud environment.

Hybrid cloud :

A Hybrid cloud combines public and private cloud capabilities, allowing flexibilities in managing workloads. VIT can benefit from hybrid solutions by :

1. Optimizing cost and Security:

Sensitive data can be stored in a private cloud, while non critical application run on a public cloud to balance cost and security.

2. Seamless Data Migration:

Hybrid cloud solutions enables smooth data migration and integration between on premises infrastructure and cloud resources.

3. Scalability for seasonal demand:

During peak admission periods (or) examinations, hybrid cloud infrastructure can dynamically scale to meet increased demand.

community cloud:

community cloud are shared among multiple organizations with common concern. VIT can approach community cloud for

1. Inter-University collaboration:

Partnering with other institution for shared research projects and academic resources can enhance innovations.

2. Shared Educational Resources:

Universities can collaborate on digital libraries, MOOC's and remote research database to optimize learning experiences.

3. Specialized & Research Network:

Creating a secure cloud environment for joint research in fields like medicine, engineering and environmental science.

By leveraging these cloud computing models, VIT can assure a more efficient, secure and a cost effective digital infrastructure.